

Chemistry, 2017 EC, entrance exam, version 2

Grade 12 · Chemistry · 2017 E.C.

Time allowed: 120 minutes

Questions: 70

Total marks: 70

Name _____ Grade _____ ID No. _____ Date _____

1. Which of the following is correct about the postulates of Thomson's experiment? (1)
- A. Cathode rays are made of positively charged particles, later called electrons.
 - B. Electrons are NOT present in all atoms of all elements.
 - C. Electrons have a very large mass compared to the atom.
 - D. Cathode rays travel in straight lines and can be deflected by electric and magnetic fields
2. Imagine you are a scientist in the 19th century and want to improve Dalton's Atomic Theory. Which of the following new experiments would best refine his ideas? (1)
- A. Designing a cathode ray tube experiment to identify subatomic particles.
 - B. Proposing that all atoms are made of fire, water, air, and earth
 - C. Conducting an experiment to prove that atoms cannot form compounds.
 - D. Showing that atoms of the same element have completely different masses.
3. The atomic mass of element A is 91 atomic mass unit. The only naturally occurring isotopes of element A are A-90 and A-94. The percent abundance in naturally occurring sample of element A are (1)
- A. 31% A-90 and 69% A-94
 - B. 75% A-90 and 25% A-94
 - C. 69% A-94 and 31% A-90
 - D. 20% A-94 and 80% A-90
4. Which of the following statements best explains the relationship between electrons' energy levels and the emission spectra of atoms? (1)
- A. Electrons absorb energy on jumping from lower to higher energy level
 - B. Electron emit energy on revolving around the nucleus
 - C. Electron absorb energy on revolving around the nucleus
 - D. The emission spectra are unrelated to electron transitions but depend on atomic size.
5. Which one of the following sets of quantum numbers could be those of the distinguishing (last) electron of Cr at s-orbital? (1)
- A. $n = 3, l = 0, m_l = 0, m_s = +1/2$
 - B. $n = 3, l = 2, m_l = -1, m_s = +1/2$
 - C. $n = 4, l = 1, m_l = 3, m_s = -1/2$
 - D. $n = 4, l = 0, m_l = 0, m_s = -1/2$
6. A radio station broadcasts at a frequency of 100 MHz. What is the wavelength of the radio waves? (Speed of light, $c = 3.00 \times 10^8$ m/s) (1)
- A. 3.0×10^{-3} m
 - B. 3.0 m
 - C. 3.0×10^6 m
 - D. 3.0×10^{-3} m
7. The photoelectric effect can be observed when: (1)
- A. Light with low intensity shined on a metal surface.
 - B. Light of sufficient frequency shining and ejects electrons from a metal
 - C. Electrons jump to higher energy levels when exposed to light
 - D. The intensity of light is increased with decreasing frequency

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8. Which of the following molecules is correctly matched with its geometry? (1)
- A. IF_5 - Trigonal pyramidal B. PCl_3 - Trigonal bipyramidal
C. BrF_3 - T-shape D. SF_4 - Square pyramidal
-
9. Which of the following is TRUE about periodic trend of elements? (1)
- A. Ionization energy increases down a group
B. Atomic size decreases across a period
C. Both atomic size and electron affinity have the same periodic trend.
D. Effective nuclear charge increases down a group.
-
10. Which of the following best describes the nature of chemical bonds? (1)
- A. A repulsive force that separates atoms
B. A temporary interaction between particles
C. An attractive force that holds atoms or ions together
D. A force that changes atomic nuclei
-
11. Which of the following condition favors the formation of an ionic bond? (1)
- A. Low ionization energy of metal and high electron affinity of non-metal
B. High ionization energy of metal and low ionization energy of non-metal
C. High electron affinity of metals and low ionization energy of non-metal
D. Large atomic size of nonmetal
-
12. The electron set geometries of NH_3 and H_2O molecules are distorted tetrahedral but bond angle in water is less than that of ammonia. Why? (1)
- A. LP-LP of H_2O repel strongly than LP-BP of NH_3 does
B. LP-BP of H_2O repel strongly than LP-BP of NH_3 does
C. Generally lone pair-lone pair repulsion is weaker than lone pair-bond pair repulsion
D. H_2O is polar and NH_3 is non-polar
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13. Which of the following is NOT true about properties of metallic bonding? (1)
- A. It gives rise to excellent electrical conductivity
B. Valence electrons are free to move throughout the structure
C. The conductivity of electric current is due to the presence of free ions
D. The strength of metallic bond affects physical properties of metals
-
14. According to Molecular orbital theory, which of the following species does NOT exist? (1)
- A. O_2^{2-} B. He_2
C. O_2 D. C_2
-
15. Which of the following molecules is the WRONG combination arrangement of the molecules with their respective molecular geometry? (1)
- A. CH_4 - AX_4 - sp^3 B. H_2O - AX_2E_2 - sp^2
C. CO_2 - AX_2 - sp D. BeCl_2 - AX_2 - sp

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- 16.** Given the following polyatomic and diatomic molecules: CH₄, NH₃, H₂S, CH₄, OF₂, NF₃, H₂O, CO₂, N₂, HF, HBr. (1)
Which one of the following sets of molecules describes the correct intermolecular force of attraction?
- A. CH₄, CCl₄, N₂, CO₂ - London force
B. CO₂, H₂O, HF, NF₃ - Hydrogen bonding
C. H₂S, CCl₄, NF₃, HBr, N₂ - Dipole-dipole force
D. H₂S, NF₃, CCl₄, HBr, N₂ - Dipole-dipole force
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- 17.** Which of the following molecule contains two pi bonds and two sigma bonds ONLY? (1)
- A. CO
B. CH₄
C. HCN
D. HO
-
- 18.** Which of the following compound would have the same electrolysis of water? (1)
- A. Concentrated NaCl (aq)
B. Diluted HCl (aq)
C. Molten Al₂O₃
D. Concentrated NaCl
-
- 19.** The pressure of a gas in a cylinder when heated to a temperature of 250 K is 1.5 atm. What is the initial temperature of the gas if its initial pressure was 1.0 atm? (1)
- A. 166.67 K
B. 300 K
C. 200 K
D. 100 K
-
- 20.** The correct increasing order of rate of diffusion of gases PH₃, O₂, CH₄, NH₃, HCl is: (1)
- A. O₂ < HCl < PH₃ < NH₃ < CH₄
B. CH₄ < NH₃ < PH₃ < O₂ < CH₄
C. HCl < PH₃ < O₂ < NH₃ < CH₄
D. PH₃ < CH₄ < NH₃ < HCl < O₂
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- 21.** Which of the following statements is TRUE about alkanes and alkenes? (1)
- A. Alkanes are unsaturated hydrocarbons, while alkenes are saturated hydrocarbons.
B. Alkanes and alkenes are both saturated hydrocarbons.
C. Alkanes are saturated hydrocarbons, while alkenes are unsaturated hydrocarbons.
D. Alkanes and alkenes are both unsaturated hydrocarbons.
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- 22.** How does the kinetic energy of gas particles compare to that of liquid and solid particles? (1)
- A. Gas particles have the least kinetic energy
B. Gas particles have the most kinetic energy
C. Gas particles have the same kinetic energy as solids
D. Gas particles have no kinetic energy
-
- 23.** Which of the following precondition of the reaction is necessarily important? (1)
- A. Proper orientation
B. High activation energy
C. Ineffective collision
D. Minimum pressure
-
- 24.** For the chemical reaction $\text{H}_2 + \text{Br}_2 \rightarrow 2\text{HBr}$, what can be said of the relative rates of consumption of the reactants when compared to the formation of product? (1)
- A. HBr will be produced at half the rate that bromine is consumed.
B. HBr will be produced at the same rate that bromine is consumed.
C. Br₂ and H₂ will be consumed at different rates.
D. HBr will be produced at twice the rate that bromine is consumed.

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- 25.** Why does powdered zinc react faster with hydrochloric acid than a solid block? (1)
- A. Powdered zinc has lower energy
 - B. Powdered zinc has a higher surface area for collision
 - C. The solid block has more atoms
 - D. Solid blocks dissolve faster
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- 26.** Which of the following is TRUE regarding the concentration of products, for a chemical reaction that is already at equilibrium, assuming no disruptions to the equilibrium? (1)
- A. The concentrations of product and reactant are equal
 - B. The rate of forward and backward reaction are equal at equilibrium
 - C. The rate of forward and backward reaction are constant but may not be equal
 - D. The concentrations of products will change continually because of reversibility.
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- 27.** If the reaction quotient Q_c has a smaller value than the related equilibrium constant, K_c : (1)
- A. The reaction is at equilibrium.
 - B. The reaction is not at equilibrium, and will make more products at the expense of reactants.
 - C. The reaction is not at equilibrium, and will make more reactants at the expense of products.
 - D. The value of K_c will decrease until it is equal to Q_c
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- 28.** What does Le Chatelier's principle state about equilibrium? (1)
- A. A system at equilibrium will always remain unchanged at any condition
 - B. If a system at equilibrium is disturbed, it will shift to oppose the disturbance.
 - C. Equilibrium is only affected by temperature changes.
 - D. The addition of a catalyst will change the equilibrium position
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- 29.** For the reaction $\text{CO(g)} + \text{H}_2\text{O(l)} \rightleftharpoons \text{CO}_2\text{(g)} + \text{H}_2\text{(g)}$, how does adding CO_2 affect equilibrium? (1)
- A. The reaction shifts toward products
 - B. The reaction shifts toward the reactants
 - C. No change occurs
 - D. The reaction stops
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- 30.** Which of the following cause the greatest shift in equilibrium to the left for the hypothetical reaction given: $\text{A(g)} \rightleftharpoons \text{B(g)} + \text{C(g)}$. (1)
- A. Increases pressure
 - B. Adding catalyst
 - C. Decrease temperature
 - D. Adding more PCl_5
-
- 31.** Which of the following correctly defines alcohol? (1)
- A. hydrocarbon containing an oxygen atom
 - B. A compound containing a hydroxyl ($-\text{OH}$) group attached to a carbon atom
 - C. A compound with an oxygen atom bonded to two alkyl groups
 - D. A compound containing carbonyl ($-\text{C}=\text{O}$) group
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- 32.** Alcohols A, B and C all have the composition $\text{C}_4\text{H}_{10}\text{O}$. Molecules of alcohol A can be oxidized to an aldehyde; molecules of alcohol B can be oxidized to a ketone; and molecules of alcohol C can be oxidized to neither an aldehyde nor a ketone. What are the correct structural formulas of molecules A, B, and C respectively? (1)
- A. A: 1-butanol, B: 2-butanol, C: 2-methyl-2-propanol
 - B. A: 2-butanol, B: 1-butanol, C: ether
 - C. A: 3-butanol, B: 2-butanol, C: 1-butanol
 - D. A: 2-methylbutanol, B: 2-butanol, C: ether

33. Which one of the following about alcohol is NOT true? (1)
- A. Dihydric and trihydric alcohols have higher boiling points than monohydric alcohols of similar molecular size
 - B. The solubility of alcohols in water increases with increasing carbon number.
 - C. The water solubility of dihydric and trihydric alcohols are higher than those of monohydric alcohols of similar molecular mass.
 - D. The boiling point of a branched isomer is lower than that of its isomeric straight-chain alcohol.
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34. Which of the following is NOT a physical property of saturated monocarboxylic acids? (1)
- A. High boiling point
 - B. Weakest intermolecular force
 - C. Soluble in water (lower members)
 - D. Oily or waxy nature (higher members)
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35. The statement accurately describes the chemical composition of fats? (1)
- A. Esters of glycerol with three saturated fatty acids
 - B. Esters of glycerol with one saturated fatty acids.
 - C. Esters of glycerol with three unsaturated fatty acids.
 - D. Esters of glycerol with two fatty acids
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36. Given the following set of compounds, alcohol and carboxylic acids. Select which one is correctly arranged in increasing order of their boiling points. (Compounds: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$, $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$, $\text{CH}_3(\text{CH}_2)_5\text{COOH}$, $\text{CH}_3\text{C}(\text{CH}_3)_2\text{COOH}$) (1)
- A. $\text{CH}_3\text{C}(\text{CH}_3)_2\text{COOH} < \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} < \text{CH}_3(\text{CH}_2)_5\text{COOH} < \text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$
 - B. $\text{CH}_3(\text{CH}_2)_5\text{COOH} < \text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3 < \text{CH}_3\text{C}(\text{CH}_3)_2\text{COOH} < \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
 - C. $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3 < \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} < \text{CH}_3(\text{CH}_2)_5\text{COOH} < \text{CH}_3\text{C}(\text{CH}_3)_2\text{COOH}$
 - D. $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3 < \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} < \text{CH}_3\text{C}(\text{CH}_3)_2\text{COOH} < \text{CH}_3(\text{CH}_2)_5\text{COOH}$
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37. Which of the following is correct about the discovery of subatomic particles? (1)
- A. The experiment of nucleus discovery is cathode ray experiment
 - B. Neutron is discovered by J.J. Thomson
 - C. Nucleus is discovered by alpha particle scattering experiment
 - D. Proton is discovered by discharge tube experiment
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38. An element has three isotopes: Isotope 1: mass = 10 amu, abundance = 2%; Isotope 2: mass = 20 amu, abundance = 95%; Isotope 3: mass = 30 amu, abundance = 3%. Which of the following is the most likely atomic mass for this element without any calculation? (1)
- A. About 23.00 amu, because it is the average mass of the isotopes.
 - B. About 20.00 amu, because the heaviest isotope is the most abundant.
 - C. About 25.00 amu, because it's the median mass of the isotopes.
 - D. About 10.00 amu, because the lightest isotope is the least abundant.
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39. How does atomic radius change as you move down a group in the periodic table, and why? (1)
- A. It decreases due to stronger nuclear attraction.
 - B. It remains constant because electron shielding has no effect.
 - C. It increases due to addition of more electron shells.
 - D. It decreases because valence electrons experience more attraction from the nucleus.
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40. Which of the following statements is true about nonpolar covalent molecules? (1)
- A. They share electrons unequally
 - B. They are always asymmetrical
 - C. They dissolve easily in water.
 - D. They have equal sharing of electrons.

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- 41.** A student needs to determine which compound has the highest boiling point from the given compounds: H_2S , CO_2 , NH_3 , or CH_4 . Based on your understanding of intermolecular forces, which option would you recommend and why? (1)
- A. H_2S , because it has the weakest London dispersion forces.
 - B. CO_2 , because it has dipole-dipole interactions.
 - C. NH_3 , because it forms strong hydrogen bonds.
 - D. CH_4 , because it has stronger dipole-dipole forces.
-
- 42.** What does the Law of Conservation of Mass state? (1)
- A. Matter can be created and destroyed in a chemical reaction.
 - B. The total mass of reactants and products are equal in a balanced chemical reaction.
 - C. The composition of a compound varies depending on how it is prepared.
 - D. The ratio of elements in a compound changes based on reaction conditions.
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- 43.** A student performs a reaction with two reactants and finds that the actual yield is significantly lower than expected. Which approach would best help them determine whether a reactant was limiting or in excess? (1)
- A. Calculate the theoretical yield based on stoichiometry and compare it with the actual yield.
 - B. Measure the temperature change of the reaction to assess reactant consumption
 - C. Weigh only one reactant before and after the reaction and assume the other is limiting.
 - D. Estimate the product yield visually to determine whether the reaction reached completion.
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- 44.** What is meant by an amphoteric species? (1)
- A. A species that can donate a proton (acid) or accept a proton (base).
 - B. A species that can only donate a proton
 - C. A species that can only accept a proton
 - D. A species that cannot donate or accept a proton.
-
- 45.** In the reaction: $\text{CH}_3\text{COOH} + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}_3\text{O}^+$, what is the acid-base conjugate pair involving the reaction? (1)
- | | |
|---|--|
| A. HCOOH and CH_3COO^- | B. CH_3COOH and H_3O^+ |
| C. CH_3COO^- and H_2O | D. H_2O and H_3O^+ |
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- 46.** A solution has a hydroxide ion concentration of $[\text{OH}^-] = 2.5 \times 10^{-3} \text{ M}$. Using $K_w = 1.0 \times 10^{-14}$, how would you determine the hydronium ion concentration $[\text{H}_3\text{O}^+]$ at 25°C ? (1)
- | | |
|--|---|
| A. $[\text{H}_3\text{O}^+] = (1.0 \times 10^{-14}) / (2.5 \times 10^{-3})$ | B. $[\text{H}_3\text{O}^+] = (2.5 \times 10^{-3}) \times (1.0 \times 10^{-14})$ |
| C. $[\text{H}_3\text{O}^+] = (1.0 \times 10^{-14}) + (2.5 \times 10^{-3})$ | D. $[\text{H}_3\text{O}^+] = (1.0 \times 10^{-14}) - (2.5 \times 10^{-3})$ |
-
- 47.** If a 0.05 M solution of $\text{Ba}(\text{OH})_2$ completely dissociates in water, what is the $[\text{OH}^-]$ and pH? (1)
- | | |
|---|---|
| A. $[\text{OH}^-] = 0.05 \text{ M}$, pH = 2 | B. $[\text{OH}^-] = 0.10 \text{ M}$, pH = 13 |
| C. $[\text{OH}^-] = 0.025 \text{ M}$, pH = 1.6 | D. $[\text{OH}^-] = 0.01 \text{ M}$, pH = 2 |
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- 48.** Two students argue about the hydroxide ion concentration in a solution with a pH of 4. Student A suggested $[\text{OH}^-] = 1.0 \times 10^{-10} \text{ M}$, while Student B suggested $[\text{OH}^-] = 1.0 \times 10^{-4} \text{ M}$. Who is correct? (1)
- A. Student A, because $\text{pOH} = 14 - \text{pH}$ and $[\text{OH}^-] = 10^{-\{\text{pOH}\}}$
 - B. Student B, because in acidic solutions $[\text{OH}^-]$ is greater than $[\text{H}^+]$
 - C. Neither, because $[\text{OH}^-]$ can't be calculated from pH
 - D. Both are correct because pH and pOH are independent values.

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49. Which of the following is the Net Effect for addition of sodium acetate (CH_3COONa) to acetic acid (CH_3COOH) solution? (1)
- A. $[\text{H}_3\text{O}^+]$ increase
B. $[\text{OH}^-]$ increase
C. pH decrease
D. ionization level of CH_3COOH increase
-
50. Which of the following is an example of a common buffer system? (1)
- A. NaCl and HCl
B. H_2CO_3 and HCO_3^-
C. H_2SO_4 and SO_4^{2-}
D. NaOH and H_2O
-
51. What is the primary function of a buffer solution? (1)
- A. To speed up chemical reactions
B. To completely neutralize acid and base
C. To resist pH changes when acids or bases are added
D. To prevent neutralization reaction
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52. If you mix acetic acid (CH_3COOH) with sodium acetate (CH_3COONa), what will be the resulting solution? (1)
- A. A strong acid solution
B. A strong base solution
C. A neutral solution
D. A buffer solution
-
53. Which of the following is the correct difference between soap and detergent? (1)
- A. Soap is better than detergent
B. Soap is made from synthetic fats or oils, while detergents are natural.
C. Soaps do not work well in hard water, but detergents work better in hard water.
D. Both soap and detergent are equally dissolve in water
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54. If a buffer solution has a pH of 7.4 and additional small amount of HCl is added, what will likely happen? (1)
- A. The pH will decrease slightly
B. The pH will increase significantly
C. The buffer will break down immediately
D. The pH will remain exactly the same
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55. Why is the bicarbonate buffer system essential in human blood? (1)
- A. It neutralizes strong acids only
B. It maintains blood pH within a narrow pH range
C. It increases oxygen levels in the blood
D. It eliminates carbon dioxide
-
56. Consider potassium phosphate (K_3PO_4) dissolving in water. Why does the solution become basic? (1)
- A. The phosphate ion (from a weak acid) hydrolyzes to produce OH^- ions.
B. The potassium ion hydrolyzes to produce OH^- ions.
C. Both ions hydrolyze equally to raise the pH.
D. Neither ion hydrolyzes.
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57. A chemist is examining the titration curve of a reaction between an acid and a base. By analyzing the curve, the chemist notices that after a certain volume of titrant has been added, the neutralization reaction is nearly complete. Based on your analysis of the titration data, which statement best defines the equivalence point? (1)
- A. Amount of titrant = Amount of analyte in a sample.
B. Amount of titrant > Amount of analyte in a sample.
C. Amount of titrant < Amount of analyte in a sample.
D. Neither of titrant nor analyte consumed

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- 58.** Which of the following is correct about oxidation-reduction reaction? (1)
- A. Oxidation reaction takes place by losing of electron
 - B. Electron is always transferred from oxidizing agent to reducing agent
 - C. Reduction takes place by increasing its oxidation number
 - D. The reaction takes place without transferring of electron.
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- 59.** Which of the following is TRUE about metallic conductors? (1)
- A. There is flow of electrical energy and transfer of matter
 - B. Electrical conduction through metals does bring about chemical transformations
 - C. Free and mobile electrons of the metallic atoms are responsible for electrical conductance
 - D. Conductivity of metal increases with increase in temperature
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- 60.** For the electrolysis of aqueous copper(II) sulfate, which of the following statements is correct? (1)
- A. Cu and O₂ are produced in mol ratio of 1:1
 - B. H₂ and O₂ are produced in mol ratio of 1:1
 - C. H₂ and O₂ are produced in mol ratio of 2:1
 - D. Cu and O₂ are produced in mol ratio of 2:1
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- 61.** What is Faraday's first law of electrolysis? (1)
- A. The amount of substance deposited is directly proportional to the voltage
 - B. The amount of substance deposited is inversely proportional to the time taken
 - C. The amount of substance deposited is inversely proportional to the charge
 - D. Amount of substance deposited is directly proportional to the charge passed through the cell
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- 62.** The process of electro refining of copper involves electrolysis. What is the primary purpose of this process in industry? (1)
- A. To add impurities to copper
 - B. To decrease the temperature of copper
 - C. To make copper a better conductor
 - D. To remove impurities from copper
-
- 63.** Given the challenges of corrosion in metal products, evaluate the effectiveness of electroplating in preventing rust formation compared to other protective methods used in the manufacturing industry. (1)
- A. Electroplating is the most effective method that forms durable protective coating on metals.
 - B. Electroplating is less effective than galvanization because it requires more maintenance.
 - C. Electroplating is ineffective in preventing rust formation at extreme temperatures.
 - D. Electroplating is only effective when metals are combined with nonmetals.
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- 64.** Which of the following statements correctly distinguishes between primary and secondary cells? (1)
- A. Primary cells can be recharged, while secondary cells cannot
 - B. Secondary cells convert chemical energy into electrical energy only once
 - C. Primary cells are for single-use, while secondary cells can be recharged
 - D. Both types of cells are rechargeable
-
- 65.** In a galvanic cell, why does oxidation occur at the anode? (1)
- A. Because the anode is always negatively charged in an electrochemical cell
 - B. Because electrons are sourced from anode and enter to cathode
 - C. Because reduction can only happen at the cathode
 - D. Because the anode gains electrons

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66. Which of the following is best describing the difference between electrolytic and galvanic cell? (1)
- A. Electrolytic cell: Requires an external power supply; Galvanic cell: Generates electrical energy
 - B. Electrolytic cell: Spontaneous ($\Delta G < 0$); Galvanic cell: Non-spontaneous ($\Delta G > 0$)
 - C. Electrolytic cell: Electrons flow from cathode to anode through an external wire; Galvanic cell: Electrons flow from anode to cathode through salt bridge.
 - D. Electrolytic cell: Oxidation takes place at anode electrode; Galvanic cell: Oxidation takes place at cathode electrode
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67. What is the primary contribution of industrial chemistry to society? (1)
- A. Reducing the need for natural resources
 - B. Producing materials essential for daily life
 - C. Reducing extent to use resources
 - D. Stopping all chemical reactions
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68. Which of the following is a renewable resource? (1)
- | | |
|-----------------|----------------|
| A. Coal | B. Petroleum |
| C. Solar energy | D. Natural gas |
-
69. Which of the following would decrease ammonia production in the Haber process? (1)
- | | |
|---|-------------------------------------|
| A. Increasing nitrogen concentration | B. Removing ammonia as it is formed |
| C. Increasing temperature significantly | D. Using an iron catalyst |
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70. Which of the following is NOT true about industrial production of sodium carbonate (Na_2CO_3)? (1)
- A. The raw materials are brine solution (NaCl), limestone (CaCO_3), ammonia and water
 - B. The byproduct of manufacturing of Na_2CO_3 in the Solvay process is CaCl_2
 - C. Burning coke in the Solvay process is required to provide the heat energy
 - D. The Solvay process involves industrial manufacturing process of Na_2SO_4

— END OF EXAMINATION —